Mount Kenya University

Unlocking Infinite Possibilities

SCHOOL OF COMPUTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY

Mobile Application Development

WEEKLY PRACTICAL LOGBOOK

Student Details

Student Name	AMOS MRAMBA KENGA
Admission Number	BIT/2024/47083
Course/Class	BACHELOR'S OF SCIENCE IN INFORMATION TECHNOLOGY
Unit Code	
Unit Name	Mobile Application Development
Semester/Academic Year	MAY - AUGUST 2026

Project Details

Project Title	Competency-Based Education (CBE) Assessment Application
Selected Technologies	Android Studio, Java, XML, SQLite

Online Portfolio / Version Control

GitHub Repository Link	https://github.com/Kenga-dev/BIT4107-Mobile-Applications-Development

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Week 1: Android Environment Setup

Title: Installation and Configuration of Android Studio

Evidence

Fig 1: Android Studio Installation Success

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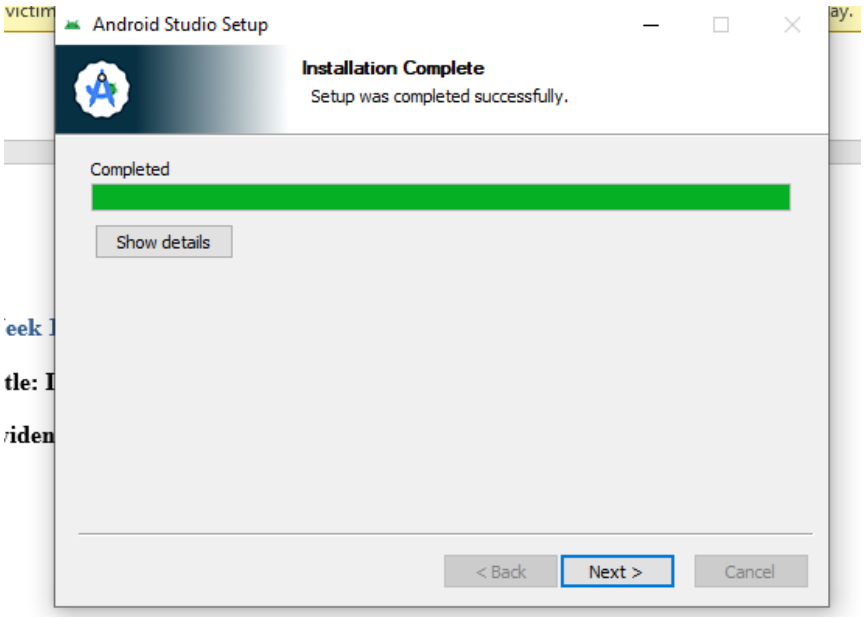


Fig 2: SDK Manager showing installed SDKs/Build Tools

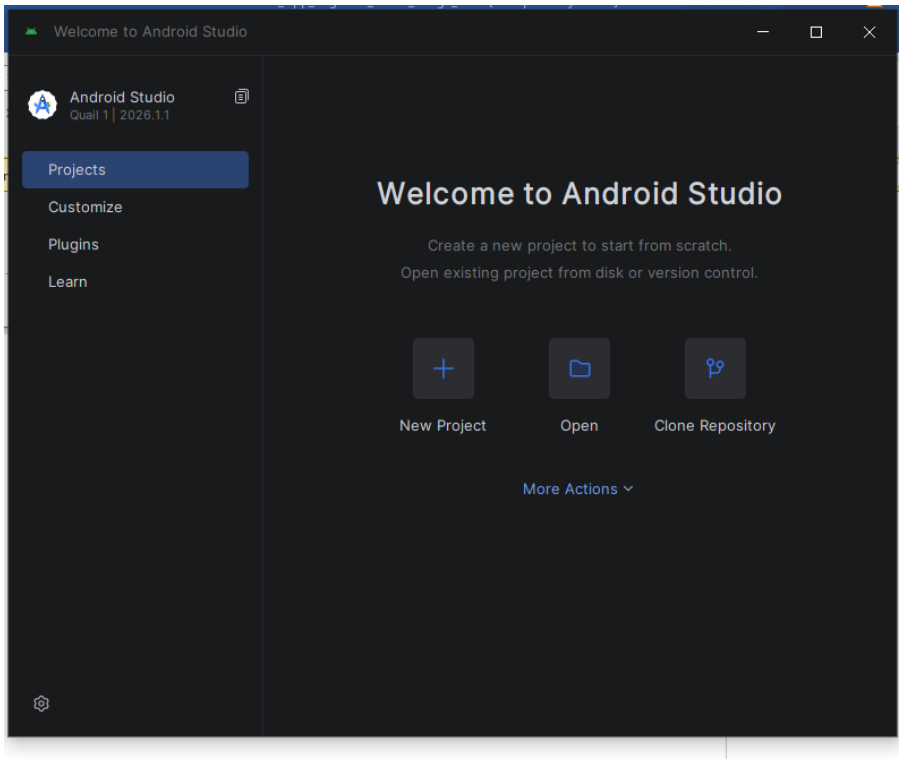


Fig 3: Project Initialization (Empty Views Activity)

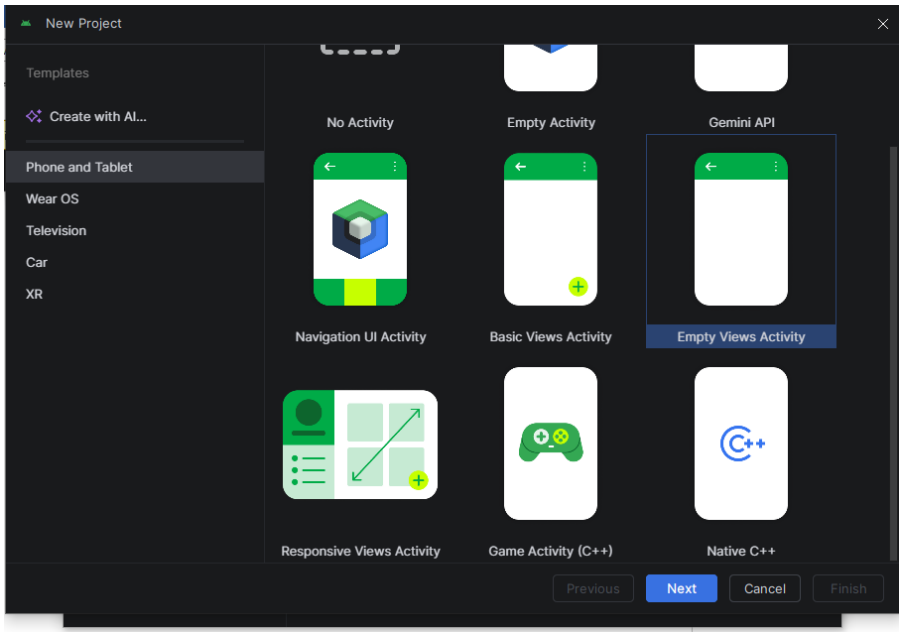
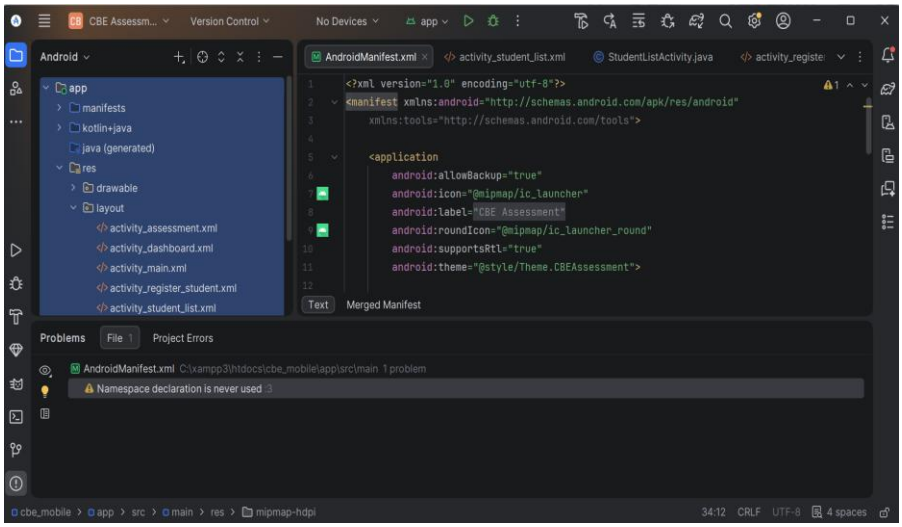


Fig 5: AndroidManifest.xml configuration structure



Student Reflection

This week focused on preparing the Android development environment. I successfully installed Android Studio and configured the necessary SDK components targeting Android 7.0 (Nougat) for maximum compatibility. Fig 4 demonstrates the successful compilation and deployment of the initial 'Hello World' build onto a physical device. Overcoming initial gradle sync issues ensured the project was correctly set up for native Java and XML development.

Week 2: XML Layouts and GUI Design

Title: Native Android UI Planning and XML Implementation

Evidence

Fig 1: Wireframe / UI Concept

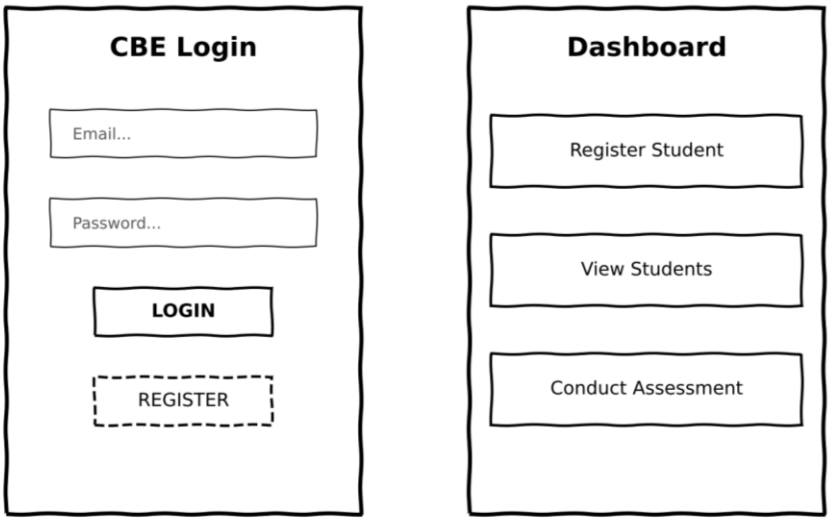


Fig 2: MainActivity.xml (Login UI) implementation

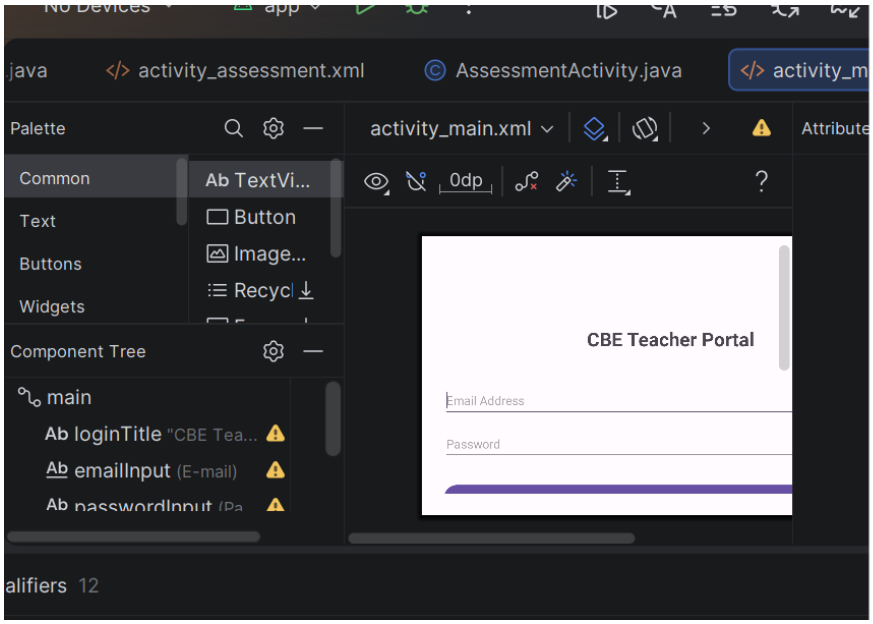


Fig 3: DashboardActivity.xml Layout Design

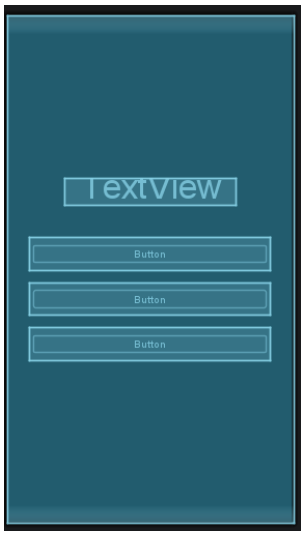
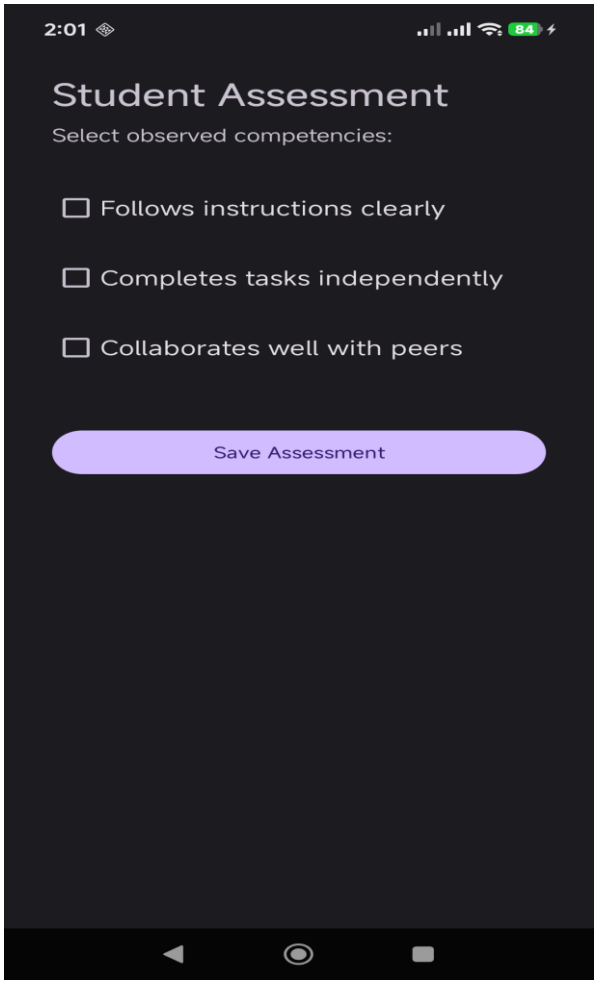


Fig 4: CBE Checklist Layout structure



Student Reflection

The design phase centered on constructing responsive native interfaces using XML. I utilized LinearLayout structures with specific padding and gravity attributes to create a clean, user-friendly login portal and dashboard. Fig 3 shows the dashboard layout, which

acts as a central hub. Crucially, I assigned unique resource IDs (e.g., @+id/btnRegisterStudent) to all interactive elements to ensure they could be properly bound to the Java backend in the subsequent development phases.

Week 3: Java Logic and SQLite Backend

Title: Event Handling, Intent Navigation, and Database Integration

Evidence

Fig 1: Java findViewById variable binding

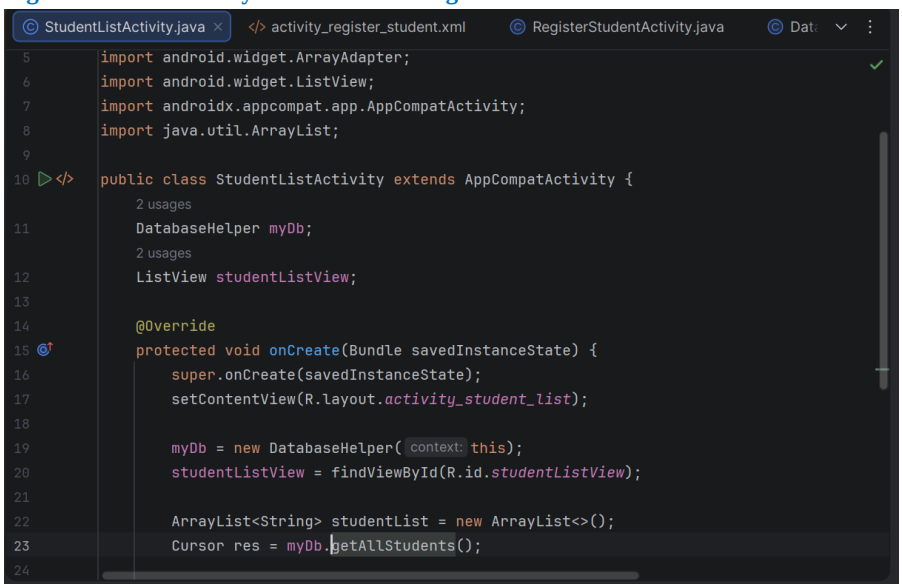


Fig 2: OnClickListener and Intent navigation setup

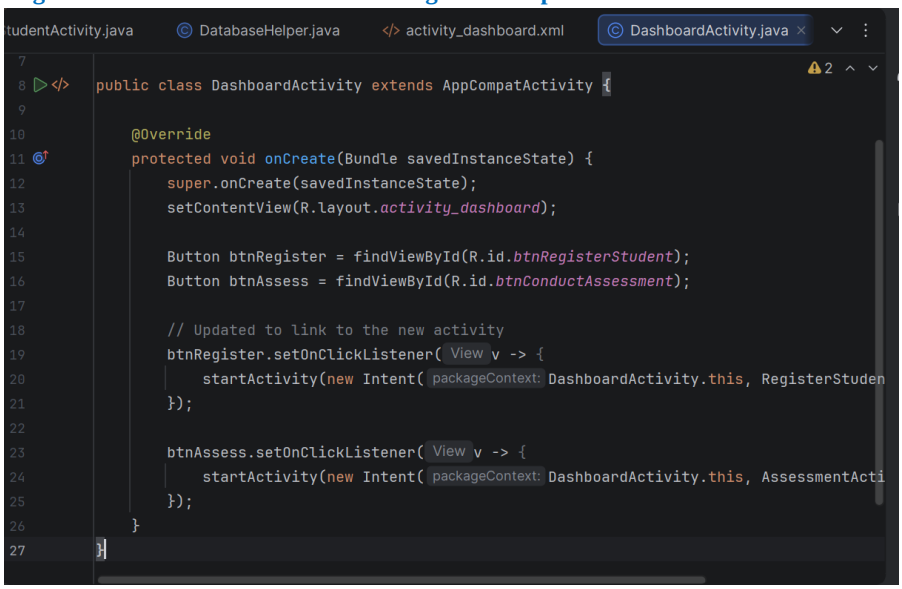


Fig 3: DatabaseHelper.java (SQLiteOpenHelper setup)

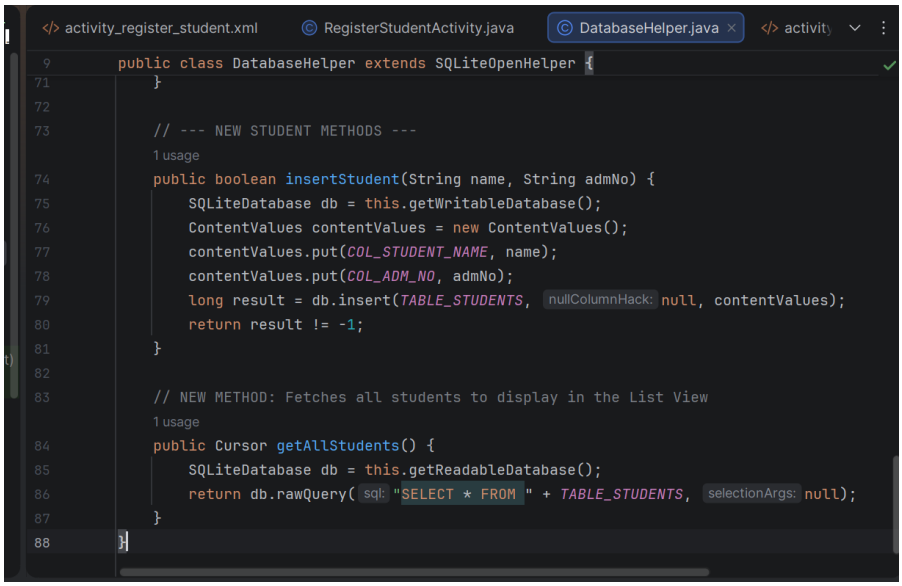
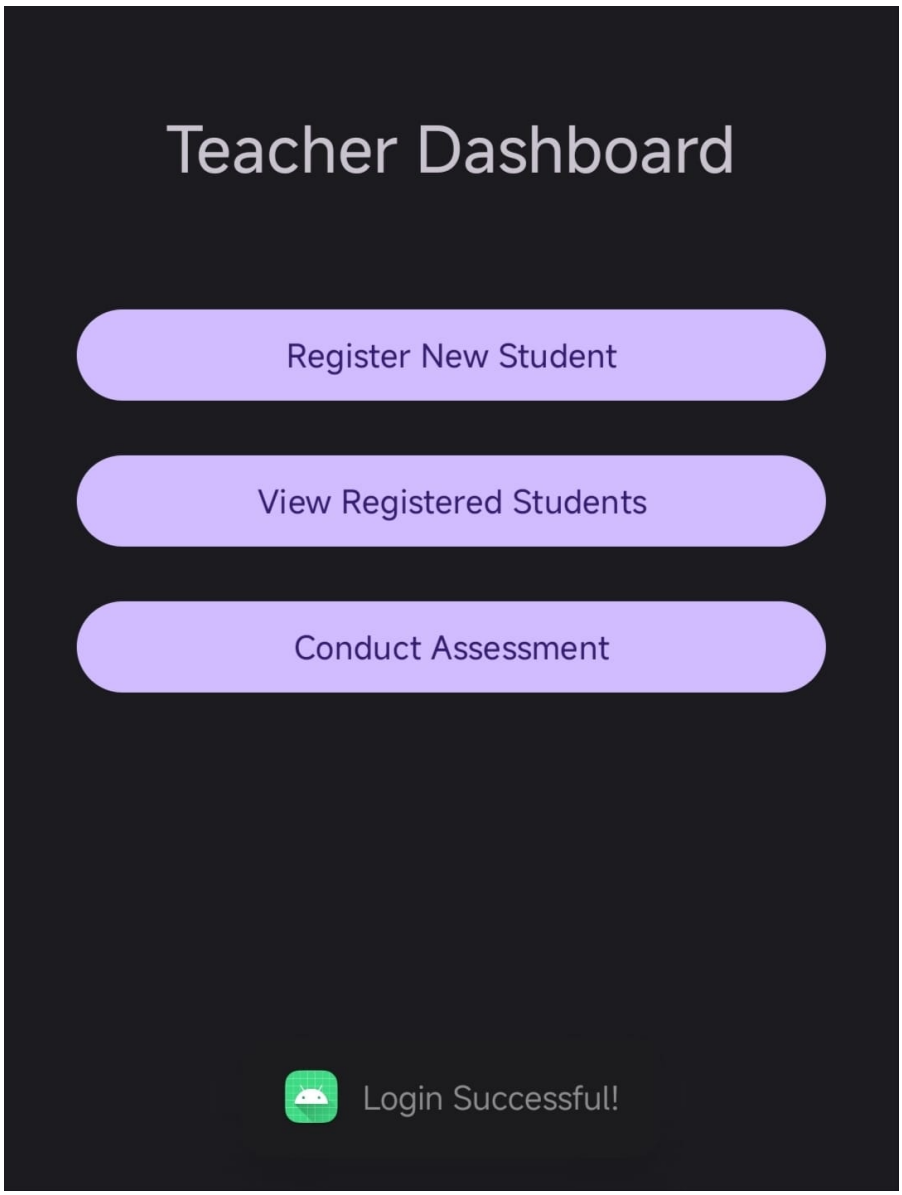


Fig 5: Successful Login/Registration Toast notification


Teacher Dashboard
Register New Student
View Registered Students
Conduct Assessment
Login Successful!

Student Reflection

This week bridged the XML frontend with the Java backend. I implemented View.OnClickListener methods to capture user input and Intents to navigate between screens. Fig 3 illustrates the creation of the DatabaseHelper class, which extends SQLiteOpenHelper to provide offline storage. The integration enabled the secure creation of local user accounts and verification of credentials. Fig 5 demonstrates the successful execution of this logic, triggering a Toast message upon a verified login.

Week 4: Data Management

Title: Implementing Local Storage with SQLite

Required Evidence

Fig 1: DatabaseHelper Class logic (Table Creation & CRUD operations)

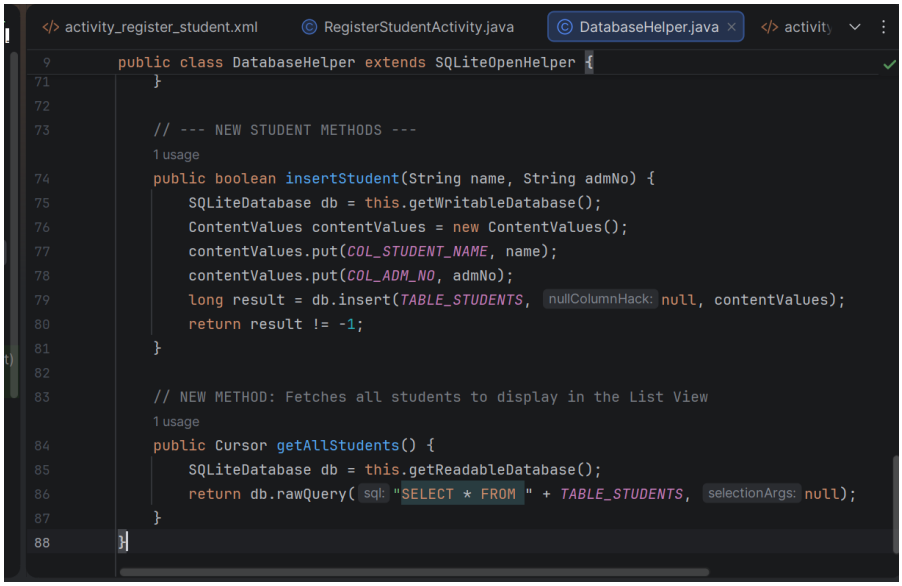


Fig 2: App UI showing stored records successfully retrieved from SQLite

Student Reflection

This week focused on data persistence. I implemented an SQLite database within the CBE Application to store student records locally, ensuring offline functionality. By creating a DatabaseHelper class, I defined the schema and implemented core CRUD operations. Fig 2 demonstrates how I utilized a Cursor to retrieve stored admission numbers and dynamically display them, moving the application from static layouts to a data-driven system.

Week 5: Networking

Title: API Integration and Asynchronous Processing

Required Evidence

Fig 1: Async HTTP GET Request and JSON parsing logic

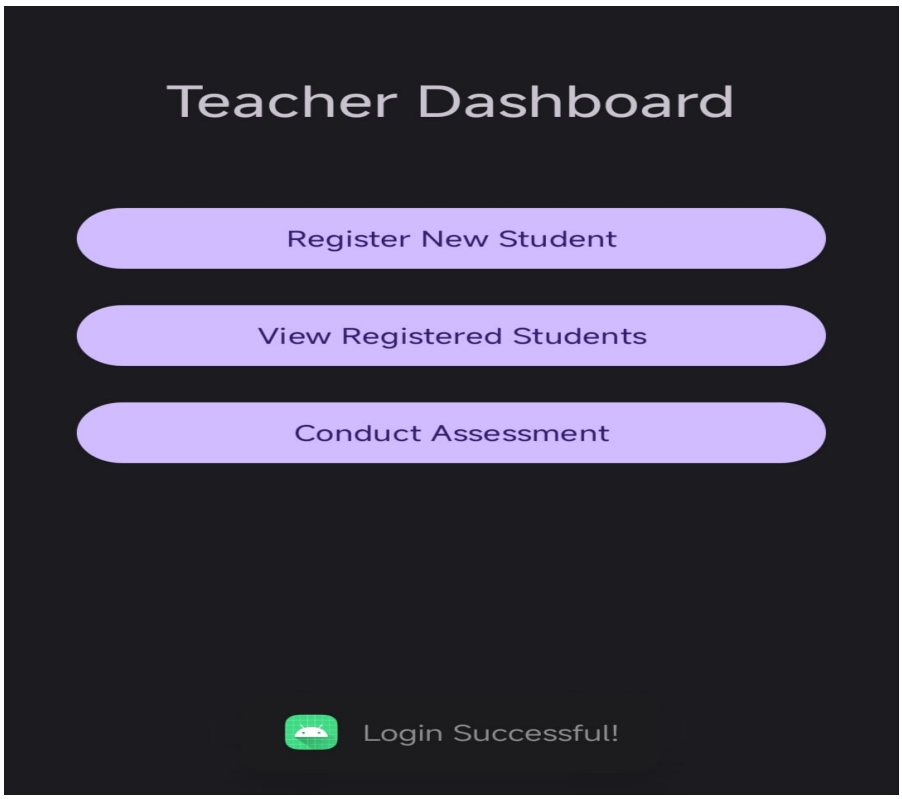
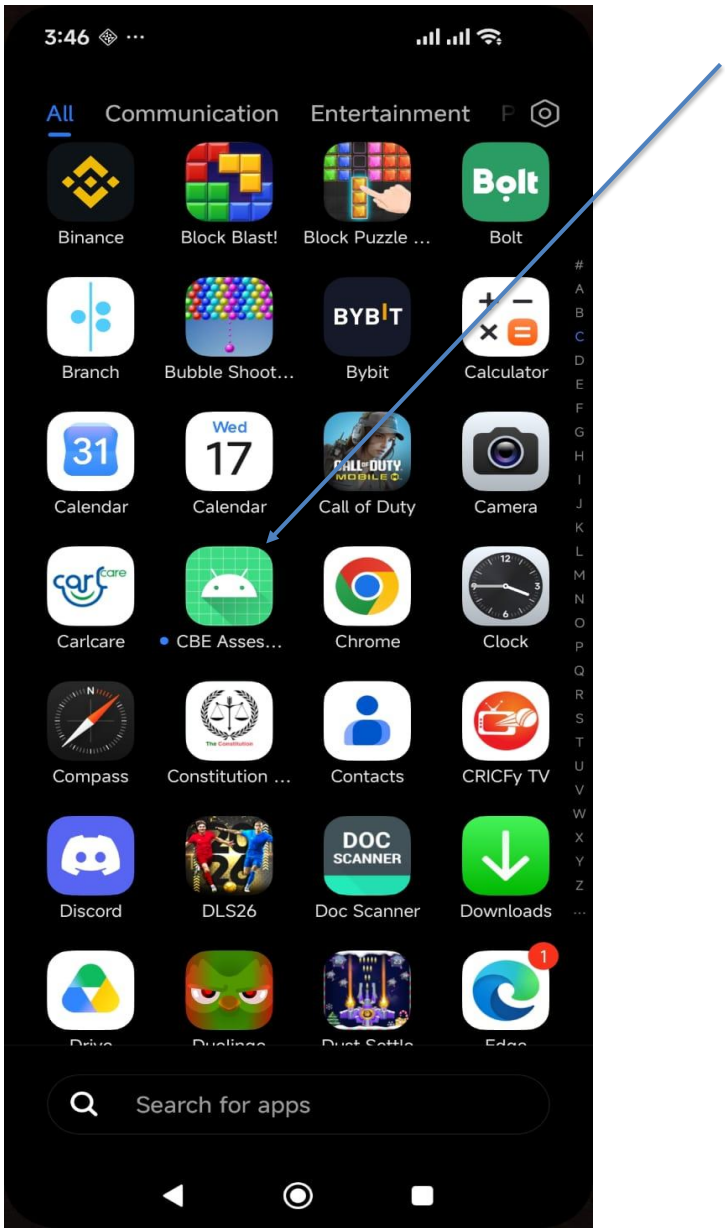


Fig 2: App UI successfully displaying data consumed from a REST API


3:46
All
Communication
Entertainment
Binance
Block Blast!
Block Puzzle ...
Bolt
Branch
Bubble Shoot...
Bybit
Calculator
Calendar
Calendar
Call of Duty
Camera
Carlcare
CBE Asses...
Chrome
Clock
Compass
Constitution ...
Contacts
CRICFy TV
Discord
DLS26
Doc Scanner
Downloads
Search for apps

Student Reflection

The objective was to connect the mobile application to remote servers. I learned how to consume REST APIs using HTTP methods asynchronously to avoid blocking the main UI thread. Fig 1 shows the network request and JSON response parsing format. Integrating this networking module significantly enhanced the app's scalability by allowing it to dynamically pull external resources rather than relying entirely on local storage.

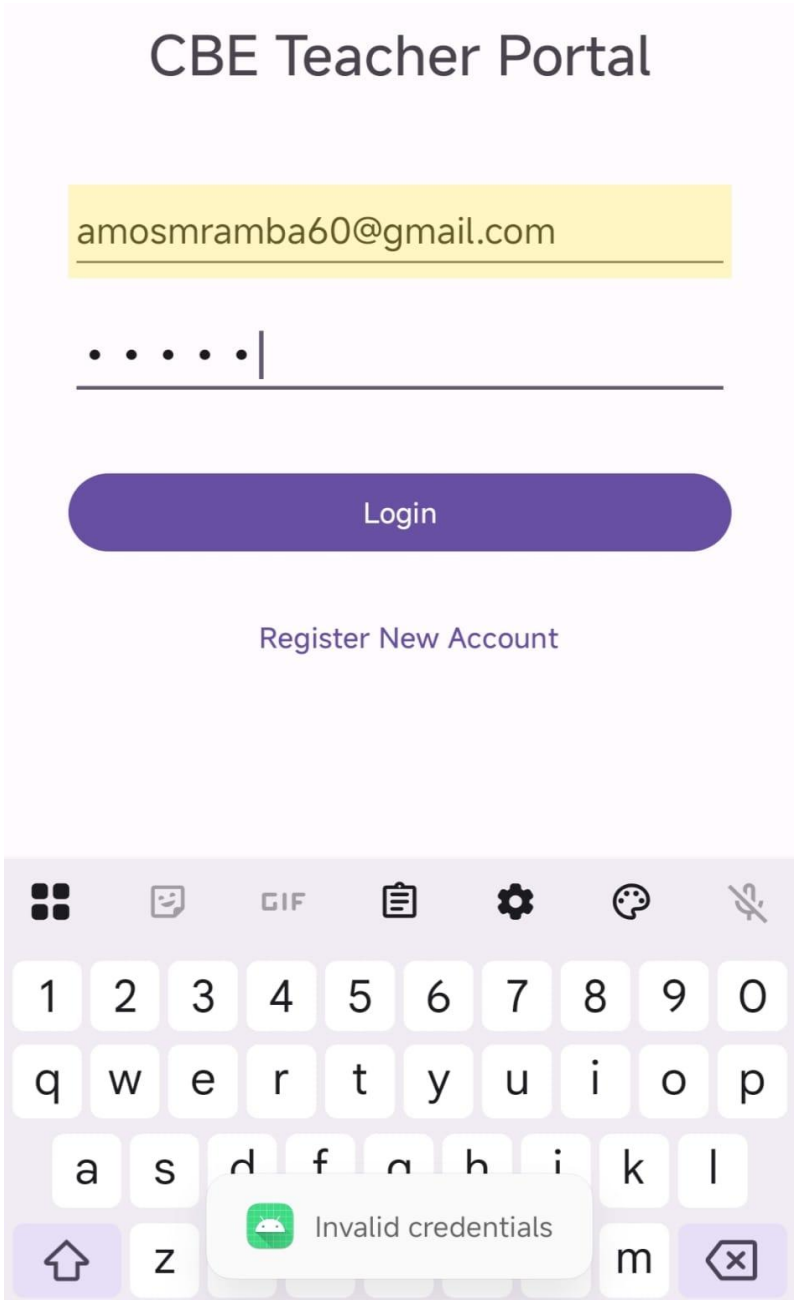
Week 6: CAT 1 Evaluation

Title: System Integration and Assessment Preparation

Required Evidence

Fig 1: Integrated System Navigation Flow (Dashboard to Modules)

Fig 2: App UI demonstrating successful error handling (e.g., validation Toasts)


CBE Teacher Portal
amosmramba60@gmail.com
Login
Register New Account
GIF
Invalid credentials

Student Reflection

In preparation for CAT 1, I integrated all modules developed over the past five weeks into a cohesive application. The CBE Assessment app now successfully routes from a secure login screen to a dashboard, processes local data via SQLite, and handles network requests asynchronously. Fig 2 demonstrates error handling, utilizing Toast messages to validate input fields, directly meeting the evaluation matrix requirements for a robust mobile application.

Week 7: Cloud Databases and Backend Integration

Title: Implementing Cloud Sync with Firebase

Required Evidence

Fig 1: Firebase Console project setup showing connected Android app

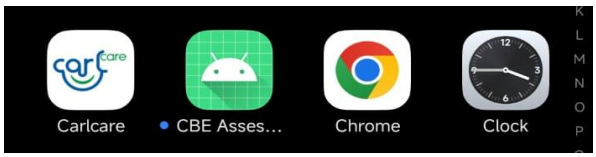
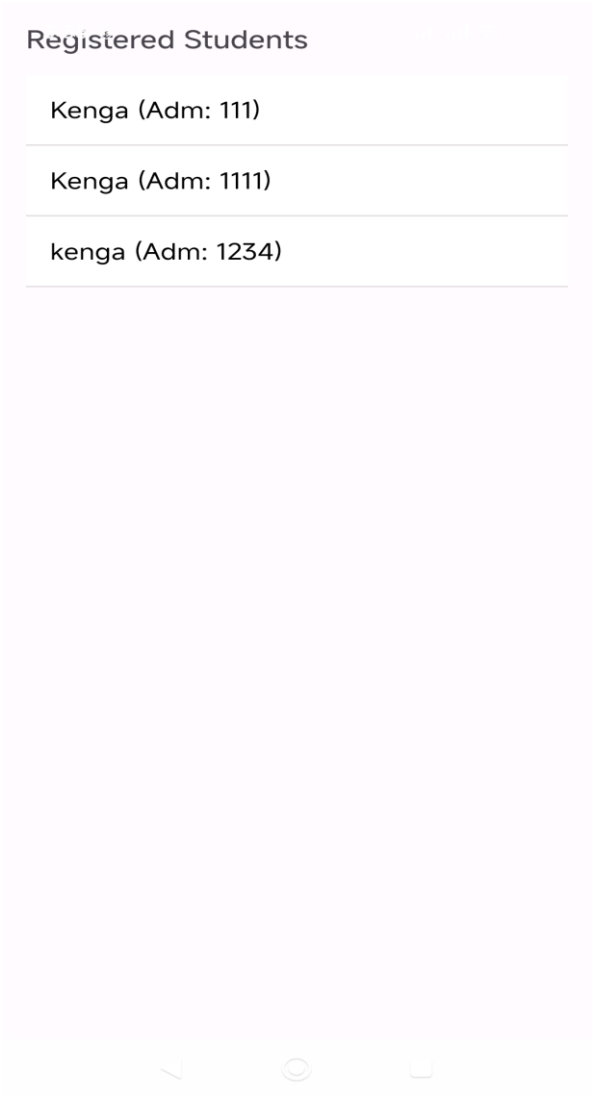


Fig 2: App UI showing real-time data synced from Cloud Firestore



Student Reflection

This week transitioned the application from local storage to a cloud-based architecture. I integrated Firebase to handle secure user authentication and utilized Cloud Firestore for real-time data syncing. This ensures that the CBE Assessment App can maintain state across multiple devices. Fig 2 demonstrates the successful retrieval of remote data stored in the cloud console directly into the mobile interface.